

NAPOLEON TORRICO

Rialto, CA • (818) 800-2052 • napoleontorricov@gmail.com
linkedin.com/in/napoleon-torrico-iv/ • github.com/nappalion

EDUCATION

California State Polytechnic University, Pomona | Pomona, CA

Aug 2021 – Dec 2023

Bachelor of Science in Computer Science, Summa Cum Laude (3.98)

Relevant coursework: Software Engineering, Mobile Application Development, Object-Oriented Design and Programming, Design and Analysis of Algorithms, Big Data Analytics, and Cloud Computing

CodePath | Remote

Certificate in Intermediate Software Engineering

Jun 2022 – Aug 2022

SKILLS

- Languages: Python, Java, Kotlin, TypeScript, JavaScript/HTML/CSS, SQL
- Tools & Frameworks: React, React Native, Expo, Zustand, React Query, SQLite, Jest, Vitest, Tailwind, Node.js, Express, Docker, Prisma, PostgreSQL, Passport.js, Firebase, Android Studio, Jetpack Compose, Figma, Git

EXPERIENCE

Senior Computer Science Instructor | Juni Learning

Oct 2021 – Nov 2025

- Curating unique advanced computer science lesson plans in the fundamentals of **Java**, **Python**, and **OO** design over Zoom for students ages 8-18 that adapt to support each student's needs and interests.

Web Development 101 Tech Fellow | CodePath

Jan 2024 – Apr 2024

- Mentored **4 groups of 5 students** each, leading them through a comprehensive curriculum including concepts in HTML, CSS, and JavaScript, resulting in a better understanding of the material and increased participation.
- Provided technical assistance on Slack for the **200-student** course while fostering an inclusive learning environment by establishing professional connections among and with students, nurturing their motivation to learn.

PROJECTS

PomoTomo | Personal Project

- Developed a full-stack Pomodoro and social productivity app using **TypeScript**, **React Native**, **Expo**, **Node.js**, and **PostgreSQL**, enabling collaborative study rooms and live chat with real-time timers via **WebSockets**.
- Engineered an offline-first architecture with local **SQLite** storage and background sync, enabling users to reliably track focus sessions without network connectivity and seamlessly synchronize data to the cloud.
- Implemented comprehensive automated testing strategies using **Jest**, **React Native Testing Library**, and **Vitest** in a monorepo architecture, achieving **85%+** code coverage.
- Designed and implemented **CI/CD** pipelines using **GitHub Actions** and **Docker**, automating testing, container builds, and deployment by publishing images to Docker Hub and pulling to production, reducing deployment time and improving release reliability.

Icebreak - Frontend Team Lead | CPP Software Engineering Association

- Led a team of 9 students using a **Node.js** API and **React Native** to create an organization management platform that incentivizes member growth.
- Adhered to **Agile**-like software development lifecycle practices, including RFCs, tests, and code reviews.
- Organized weekly standups and collaborated with team members to complete development sprints.

Pallet Configuration | Personal Project

- Designed a pallet configuration algorithm inspired by Tetris-solving AI in **JavaScript**, using **React Native** for the UI and **Firebase** for storage.
- Achieved an average runtime reduction of **85%** from **6.52 to 0.98 seconds** for **100** different boxes by employing array optimization and caching techniques.
- Increased pallet load efficiency by **15%** with an average of **6.22** more boxes per pallet by comparing algorithms for static and dynamic box rotation.
- Employed the **Three.js** library to generate and render 3D graphics for pallet layouts dynamically.